🟦 UNIT–I

(12 IMP QUESTIONS – 8 MARK WRITE SUPPORT)

Q1. Explain components of CPU

How to write (8 marks):

- Define CPU as central processing unit and brain of computer
- Explain **ALU** with examples (addition, logical ops, hashing)
- Explain **Control Unit** (fetch–decode–execute cycle)
- Explain **Registers** (PC, IR, MAR, MBR, ACC) with role
- Explain **Cache memory** (L1/L2/L3) and speed
- Explain **System bus** (data, address, control)
- Add **forensic relevance** (memory analysis, malware execution)
- Short conclusion

Q2. Explain Process Control Block (PCB)

- Define PCB and its role in multitasking OS
- Process identification (PID, PPID, user ID)
- Processor state info (registers, PC, stack pointer)
- Process state (new, ready, running, waiting, terminated)
- Memory management info (page tables, segments)
- I/O status (open files, devices)
- Forensic use (identify malicious running processes)
- Conclusion

Q3. Explain VBR and GPT

VBR

- First sector of partition
- Bootstrapping code
- BIOS Parameter Block (file system details)
- NTFS vs FAT differences

GPT

- Why GPT replaced MBR
- Protective MBR
- Primary GPT header
- Partition entry array
- Backup GPT header
- Forensic importance (partition recovery)
- Conclusion

Q4. Number system conversion

- Importance of number systems in computers
- Stepwise decimal → hexadecimal
- Stepwise decimal → binary
- Explain remainder method
- Show final answer
- Forensic importance (hex editors, disk sectors)
- Conclusion

Q5. Little Endian vs Big Endian

- Define endianness
- Explain little endian with example
- Explain big endian with example

- Architectures using each
- Importance in memory & disk analysis
- Effect on timestamps & addresses
- Forensic example
- Conclusion

Q6. Basic Disk vs Dynamic Disk

Basic Disk

- Uses MBR/GPT
- Partition types
- Operations supported

Dynamic Disk

- Uses LDM database
- Volume types (simple, spanned, striped, mirrored)
- RAID support
- Forensic challenges
- Comparison + conclusion

Q7. Locard's Principle (Digital)

- Define Locard's principle
- Digital interpretation
- File system traces
- Registry traces
- Logs & network traces
- Memory artifacts
- Anti-forensics detection

- Conclusion

Q8. Master Boot Record (MBR)

- Define MBR
- Location (LBA 0)
- Structure (boot code, partition table, signature)
- Partition table entries
- Limitations of MBR
- Bootkits relevance
- Diagram mention
- Conclusion

Q9. HDD vs SSD

- HDD structure (platters, sectors)
- SSD structure (NAND, controller)
- Data deletion behavior
- TRIM effect
- Forensic recovery difficulty
- Comparison
- Conclusion

Q10. Boot Process

- BIOS/UEFI role
- MBR/GPT loading
- Bootloader execution
- Kernel loading

- User space initialization
- Malware at boot level
- Forensic importance
- Conclusion

Q11. Process vs Thread

- Define process
- Define thread
- Resource sharing
- Execution behavior
- Scheduling
- Malware perspective
- Comparison
- Conclusion

Q12. Memory Management

- Physical vs virtual memory
- Paging concept
- Swapping
- Page tables
- RAM artifacts
- Forensic importance
- Conclusion

🟦 **UNIT–II**

(12 IMP QUESTIONS – 8 MARK WRITE SUPPORT)

Q1. Digital Forensics

- Definition (scientific + legal)
- Scope of digital evidence
- Devices involved
- Importance in cybercrime
- Legal admissibility
- Tools & techniques
- Conclusion

Q2. Digital Forensics Investigation Process

- Importance of structured approach
- Identification
- Preservation
- Collection
- Examination
- Analysis
- Presentation
- Conclusion

Q3. Locard's Principle

- Principle definition
- File artifacts
- Registry artifacts
- Log artifacts
- Network artifacts

- RAM artifacts
- Anti-forensic traces
- Conclusion

Q4. Handling Digital Crime Scene

- Securing scene
- Isolating devices
- Documentation
- Volatile data collection
- Proper shutdown
- Packaging & transport
- Chain of custody
- Conclusion

Q5. Electronic Evidence Types

- Computer evidence
- Mobile evidence
- Network evidence
- Cloud evidence
- IoT evidence
- Removable media
- Importance
- Conclusion

Q6. Chain of Custody

- Definition

- Elements
- Transfer records
- Storage
- Legal requirement
- Consequences of failure
- Example
- Conclusion

Q7. Write Blockers

- Definition
- Purpose
- Hardware vs software
- How it works
- Forensic standards
- Risks without write blocker
- Conclusion

Q8. Imaging Techniques

- Forensic imaging definition
- Bit-stream imaging
- Logical imaging
- Live imaging
- Targeted imaging
- Hash validation
- Conclusion

Q9. Evidence Integrity & Hashing

- Evidence integrity definition
- Hash algorithms
- Pre & post imaging hash
- Integrity verification
- Court acceptance
- Conclusion

Q10. Volatile vs Non-Volatile Evidence

- Volatile data definition
- Examples (RAM)
- Non-volatile data
- Order of volatility
- Forensic importance
- Conclusion

Q11. Legal Issues in Digital Forensics

- Search warrants
- Consent
- Jurisdiction
- Privacy laws
- Scope limitation
- Evidence admissibility
- Conclusion

Q12. Documentation

- Evidence forms
- Investigator notes
- Scene photos
- Case report
- Importance
- Legal value
- Conclusion

🟦 UNIT–III

(12 IMP QUESTIONS – 8 MARK WRITE SUPPORT)

Q1. Windows OS Architecture

- User mode
- Kernel mode
- Executive services
- Kernel
- HAL
- Drivers
- Forensic relevance
- Conclusion

Q2. FAT File System

- FAT overview
- FAT12/16/32
- Boot sector
- FAT table

- Root directory
- Deletion behavior
- Forensic importance
- Conclusion

Q3. NTFS File System

- NTFS overview
- Boot sector
- MFT
- System files
- Journaling
- Metadata
- Forensic importance
- Conclusion

Q4. Master File Table (MFT)

- MFT definition
- Entry structure
- Attributes
- Resident vs non-resident
- Deleted entries
- Timeline
- Conclusion

Q5. FAT vs NTFS

- Security

- Metadata
- Recovery
- Journaling
- Forensic comparison
- Conclusion

Q6. Recreating Partitions

- FAT reconstruction
- NTFS reconstruction
- Tools
- Importance
- Anti-forensics
- Conclusion

Q7. Windows Registry

- Definition
- Structure
- Hives
- Value types
- Storage location
- Forensic importance
- Conclusion

Q8. Registry Hives

- SYSTEM
- SOFTWARE

- SAM
- SECURITY
- NTUSER.DAT
- Forensic usage
- Conclusion

Q9. Registry Artifacts

- RecentDocs
- RunMRU
- UserAssist
- USBSTOR
- ShellBags
- Timeline use
- Conclusion

Q10. Windows Event Logs

- Event log types
- Storage location
- Log structure
- Event levels
- Forensic value
- Conclusion

Q11. Event IDs

- Authentication events
- System events

- Application events
- Incident detection
- Conclusion

Q12. Prefetch Files

- Purpose
- Stored data
- Execution count
- Malware detection
- Forensic relevance
- Conclusion

🟦 UNIT–III (WINDOWS FORENSICS)

Q1. Explain Windows Log Sources and their Forensic Importance (8 Marks)

Write structure:

- Definition of Windows log sources as system-generated records
- Purpose of logs as digital footprints
- Windows Event Logs
  - .evt / .evtx format
  - Storage location
- Security Logs
  - Login success/failure
  - Privilege escalation
- System Logs

  - Boot/shutdown
  - Driver loading
- Application Logs
  - Antivirus, browser, email logs
- Registry-based logs
  - USB usage
  - Program execution history
- Forensic importance
  - Timeline reconstruction
  - Intrusion detection
  - Court-admissible evidence
- Conclusion

Q2. Explain Windows Registry Files and their Forensic Significance (8 Marks)

- Define Windows Registry
- Explain hierarchical structure (hives, keys, values)
- SYSTEM hive
  - Hardware, drivers, boot config
- SOFTWARE hive
  - Installed programs
- SAM
  - User accounts, password hashes
- SECURITY
  - Audit and security policies

- NTUSER.DAT
    - User activity and preferences
- Registry file locations
- Forensic significance
    - User activity tracking
    - Malware persistence detection
    - Timeline analysis
- Conclusion

Q3. Explain Windows Events and Event IDs in Forensic Analysis (8 Marks)

- Define Windows events
- Meaning of Event IDs
- Event storage in Event Logs
- Event types
    - Information
    - Warning
    - Error
    - Critical
- Common forensic event categories
    - Logon/logoff
    - Account management
    - System boot/shutdown
    - Application execution
- Forensic importance

    - Unauthorized access detection
    - Incident time identification
    - User attribution
- Conclusion

Q4. Explain Different Types of Windows Logs (8 Marks)

- Define Windows logs
- System Logs
    - OS operations
    - Hardware & driver issues
- Security Logs
    - Authentication
    - Privilege usage
- Application Logs
    - Software errors
    - Malware behavior
- Setup Logs
    - OS install & updates
- Forwarded Events
    - Centralized enterprise logging
- Log storage location
- Forensic importance
    - Misuse detection
    - Timeline creation

- Conclusion

Q5. Explain FAT File System and its Forensic Importance (8 Marks)

- Definition of FAT file system
- Cross-platform usage (USB, SD cards)
- Types
    - FAT12
    - FAT16
    - FAT32
- Common FAT structure
    - Boot sector (VBR)
    - FAT table
    - Root directory
    - Data area
- File deletion behavior (0xE5)
- Slack space concept
- Forensic importance
    - Easy deleted file recovery
    - Common in data theft cases
- Conclusion

Q6. Explain NTFS File System and its Forensic Significance (8 Marks)

- Definition of NTFS
- NTFS structure

    - Boot sector
    - Master File Table (MFT)
    - System files ($LogFile, $Bitmap, $MFTMirr)
    - Data area
- Role of MFT
- Journaling feature
- Multiple timestamps (MAC)
- Forensic significance
    - Deleted file recovery
    - Timeline reconstruction
    - Anti-forensics detection
- Conclusion

🟦 UNIT–IV (LINUX FORENSICS)

Q1. Explain Linux Log Sources and their Forensic Importance (8 Marks)

- Define Linux log sources
- Default log directory (/var/log)
- syslog / messages
    - General system activity
- auth.log / secure
    - Login, SSH, sudo
- kern.log
    - Kernel & driver events

- dmesg
  - Hardware & USB connections
- boot.log
  - Startup processes
- Binary logs
  - wtmp, btmp, lastlog
- Forensic importance
  - Intrusion detection
  - Privilege escalation analysis
- Conclusion

Q2. Explain Recreation of EXT Partitions in Linux Forensics (8 Marks)

- Define EXT partition recreation
- EXT variants (EXT2, EXT3, EXT4)
- Important EXT components
  - Superblock
  - Block groups
  - Inode tables
  - Data blocks
  - Journal
- Recreation steps
  - Identify EXT signatures
  - Superblock analysis
  - Block group & inode reconstruction

    - Journal analysis
    - Rebuild partition table
- Tools used (TestDisk, Sleuth Kit)
- Forensic importance
    - Recover deleted partitions
    - Detect anti-forensics
- Conclusion

🟦 UNIT–V (macOS FORENSICS)

Q1. Explain Log Types of macOS and their Forensic Importance (8 Marks)

- Define macOS logging system
- UNIX-based logging architecture
- System Logs
    - Startup, kernel, hardware
- Security Logs
    - Login, sudo, authentication
- Application Logs
    - Safari, Chrome, Mail
- Install Logs
    - Software & update installation
- Unified Logging System
    - log show, log collect
- Forensic importance

    - Timeline reconstruction
    - Intrusion detection
- Conclusion

Q2. Explain Basics of HFS and HFS+ File Systems with Forensic Importance (8 Marks)

HFS

- Legacy Apple file system
- B-tree based structure
- Components:
    - Volume header
    - Catalog file
    - Extent overflow file
- No journaling
- Forensic relevance (legacy recovery)

HFS+

- Improved HFS
- Unicode filenames
- Journaling support
- Metadata-rich structure
- Forensic importance
    - Deleted file recovery
    - Timeline analysis
- Conclusion

Q3. Explain Basics of APFS File System and its Forensic Importance (8 Marks)

- Define APFS
- Designed for SSDs
- Key features
  - Containers & volumes
  - Copy-on-Write
  - Snapshots
  - Encryption
- APFS components
  - Container
  - Volumes
  - Object map
  - Snapshots
- Forensic importance
  - Snapshot-based recovery
  - Metadata protection
- Challenges
  - Encryption
  - CoW complicates carving
- Conclusion

✅ FINAL EXAM TIP (VERY IMPORTANT)

For EVERY 8-MARK ANSWER, follow this fixed pattern:

1. Definition – 4 lines

2. Core explanation – 6 to 8 bullet points
3. Forensic importance – separate heading
4. 2-line conclusion

➡️ This alone is enough to score 7–8 marks consistently.
